RISC-V RV32I Base Instruction Set

[1] RISC-V RV32I Instruction Set Listing, Table 25.2, page 130 (pdf reader page 146/236)

RISC-V Instruction Set Manual, Vol I, Unprivileged ISA Document, Version 20190608-Base-Ratified, Editors: Andrew Waterman, Krste Asanovic, Jun 8, 2019

vicilogic RV32I Base Instruction Set (this document)

Interactive RISC-V RV32I Base Instruction Set Table

ID (0x)	ID (0d)	instruction mnemonic	description	instruction(31:0) format						Type	instruction syntax
				31			12	11 7	6 opcode 0		
1	1	LUI	load upper immediate	imm[31:12]				rd	0110111	U	lui rd, imm
2	2	AUIPC	add upper immed to PC	imm[31:12]				rd	0010111	U	auipc rd, imm
		Jump and link									
3	3	JAL	jump and link	imm[20¦10:1¦11¦19:12]				rd	1101111	J	jal rd, imm
		Branch instructions		31 25	24 20	19 15	f3 14 12	11 7	6 0		
4	4	BEQ	branch if equal	imm[12¦10:5]	rs2	rs1	000	imm[4:1¦11]	1100011	B	beq rs1, rs2, imm
5	5	BNE	branch if not equal	imm[12¦10:5]	rs2	rs1	001	imm[4:1¦11]	1100011	B	bne rs1, rs2, imm
6	6	BLT	branch if less than (lt)	imm[12¦10:5]	rs2	rs1	100	imm[4:1¦11]	1100011	B	blt rs1, rs2, imm
7	7	BGE	branch if greater than (gt)	imm[12¦10:5]	rs2	rs1	101	imm[4:1¦11]	1100011	B	bge rs1, rs2, imm
8	8	BLTU	branch if lt than or eq unsgn	imm[12¦10:5]	rs2	rs1	110	imm[4:1¦11]	1100011	B	bltu rs1, rs2, imm
9	9	BGEU	branch if gt than unsgn	imm[12¦10:5]	rs2	rs1	111	imm[4:1¦11]	1100011	B	bgeu rs1, rs2, imm
		Jump and link (registered)		31	20	19 15	14 12	11 7	6 0		
0xA	10	JALR	jump & link register	imm[11:0]		rs1	000	rd	1100111	I	jalr rd, imm(rs1)
		Memory Load Instructions									
0xB	11	LB	load byte	imm[11:0]		rs1	000	rd	0000011	I	lb rd, imm(rs1)
0xC	12	LH	load halfword	imm[11:0]		rs1	001	rd	0000011	I	lw rd, imm(rs1)
0xD	13	LW	load word	imm[11:0]		rs1	010	rd	0000011	I	lw rd, imm(rs1)
0xE	14	LBU	load byte unsigned	imm[11:0]		rs1	100	rd	0000011	I	lbu rd, imm(rs1)
0xF	15	LHU	load halfword unsigned	imm[11:0]		rs1	101	rd	0000011	I	lhu rd, imm(rs1)
		Integer register-immediate instructions									
0x10	16	ADDI	add immediate	imm[11:0]		rs1	000	rd	0010011	I	addi rd, rs1, imm
0x11	17	SLTI	set less than immediate	imm[11:0]		rs1	010	rd	0010011	I	slti rd, rs1, imm
0x12	18	SLTIU	set less than immed unsgn	imm[11:0]		rs1	011	rd	0010011	I	sltiu rd, rs1, imm
0x13	19	XORI	xor immediate	imm[11:0]		rs1	100	rd	0010011	I	xori rd, rs1, imm
0x14	20	ORI	or immediate	imm[11:0]		rs1	110	rd	0010011	I	ori rd, rs1, imm
0x15	21	ANDI	and immediate	imm[11:0]		rs1	111	rd	0010011	I	andi rd, rs1, imm
		Memory Store Instructions		31 25	24 20	19 15	14 12	11 7	6 0		
0x16	22	SB	store byte	imm[11:5]	rs2	rs1	000	imm[4:0]	0100011	S	sb rs2, imm(rs1)
0x17	23	SH	store halfword	imm[11:5]	rs2	rs1	001	imm[4:0]	0100011	S	sh rs2, imm(rs1)
0x18	24	SW	store word	imm[11:5]	rs2	rs1	010	imm[4:0]	0100011	S	sw rs2, imm(rs1)
		Constant shift Instructions		f7							
0x19	25	SLLI	constant-shift left	0000000	shamt	rs1	001	rd	0010011	I	slli rd, rs1, shamt
0x1A	26	SRLI	constant-shift right	0000000	shamt	rs1	101	rd	0010011	I	srli rd, rs1, shamt
0x1B	27	SRAI	constant-shift right arithmetic	0100000	shamt	rs1	101	rd	0010011	I	srai rd, rs1, shamt
		Integer register-register instructions									
0x1C	28	ADD	add	0000000	rs2	rs1	000	rd	0110011	R	add rd, rs1, rs2
0x1D	29	SUB	subtract	0100000	rs2	rs1	000	rd	0110011	R	sub rd, rs1, rs2
0x1E	30	SLL	register-shift left	0000000	rs2	rs1	001	rd	0110011	R	sll rd, rs1, rs2
0x1F	31	SLT	set less than	0000000	rs2	rs1	010	rd	0110011	R	slt rd, rs1, rs2
0x20	32	SLTU	set less than unsigned	0000000	rs2	rs1	011	rd	0110011	R	sltu rd, rs1, rs2
0x21	33	XOR	xor	0000000	rs2	rs1	100	rd	0110011	R	xor rd, rs1, rs2
0x22	34	SRL	register-shift right (logical)	0000000	rs2	rs1	101	rd	0110011	R	srl rd, rs1, rs2
0x23	35	SRA	register-shift right (arithmetic)	0100000	rs2	rs1	101	rd	0110011	R	sra rd, rs1, rs2
0x24	36	OR	or	0000000	rs2	rs1	110	rd	0110011	R	or rd, rs1, rs2
0x25	37	AND	and	0000000	rs2	rs1	111	rd	0110011	R	and rd, rs1, rs2
				31	20	19 15	14 12	11 7	6 0		
0x26	38	FENCE		fm pred	succ	rs1	000	rd	0001111		
0x27	39	ECALL		000000000000		00000	000	00000	1110011		
0x28	40	EBREAK		000000000001		00000	000	00000	1110011		